Xiaoling Hu

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Website: `https://huxiaoling.github.io/` *Google Scholar*: 1400+ citations

Current Position

- **Harvard Medical School, Harvard University, USA** Aug. 2023 - Present
  Postdoctoral Research Fellow
  - Hosted by Prof. Juan Eugenio Iglesias and Prof. Bruce Fischl

Research Interests

My research develops principled machine learning methods that make models structurally correct, reliable, and interpretable, with applications spanning computer vision, scientific imaging, and foundation models. In particular:

- **Topological Deep Learning**
- **Trustworthy Machine Learning**
- **Multimodal AI and Generative AI (GenAI)**
- **VLM/LLM Reasoning, Efficiency and Confidence**

Education

- **Department of Computer Science, Stony Brook University, USA**
  Jan. 2018 - Jun. 2023
  Ph.D. in Computer Science
  - Advisor: Chao Chen
  – Thesis: Learning Topological Representations for Deep Image Understanding

- **Department of Electronic Engineering, Tsinghua University, China**
  Sep. 2014 - Jun. 2017
  Master of Engineering

- **Department of Electronic Engineering, Huazhong University of Science and Technology, China** Sep. 2010 - Jun. 2014
  Bachelor of Science

Selected Honors and Awards

- Top Area Chair, NeurIPS 2026
- Oral, CVPR (Top 1%) 2025
- Catacosinos Fellowship, Stony Brook University 2023
- Spotlight, ICLR (Top 6%) 2023
- Oral, ECCV (Top 3%) 2022
- Spotlight, ICLR (Top 6%) 2021

Professional Service

Area Chair

- Conference on Neural Information Processing Systems (NeurIPS) 2025, 2026
- International Conference on Learning Representations (ICLR) 2027
- International Conference on Machine Learning (ICML) 2027
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026, 2027
- International Conference on Artificial Intelligence and Statistics (AISTATS) 2026, 2027

- AAAI Conference on Artificial Intelligence (AAAI) 2027
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2027
- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2025, 2026

Reviewer

- *Conferences*: NeurIPS, ICLR, ICML, CVPR, ICCV, ECCV, AAAI, AISTATS, WACV, LoG, MICCAI
- *Journals*: TPAMI, TMLR, IEEE TMI, Medical Image Analysis (MedIA)

Selected Publications

(* indicates equal contribution, __ denotes students I (co-)mentored on that paper, ‡ denotes (co)-senior supervision)

[1] **LoC-Path: Learning to Compress for Pathology Multimodal Large Language Models**
Qingqiao Hu, Weimin Lyu, Meilong Xu, Kehan Qi, **Xiaoling Hu**, Saumya Gupta, Jiawei Zhou, Chao Chen
*IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**)*, 2027

[2] **Uncertainty Estimation for Pretrained Medical Image Registration Models via Transformation Equivariance**
Lin Tian, **Xiaoling Hu**, Juan Eugenio Iglesias
*Conference on Neural Information Processing Systems (**NeurIPS**)*, 2026

[3] ***RankByGene*: Gene-Guided Histopathology Representation Learning Through Cross-Modal Ranking Consistency**
Wentao Huang, Meilong Xu, **Xiaoling Hu**, Shahira Abousamra, Aniruddha Ganguly, Saarthak Kapse, Alisa Yurovsky, Prateek Prasanna, Tahsin Kurc, Joel Saltz, Michael L. Miller, Chao Chen
*IEEE Transactions on Medical Imaging (**TMI**)*, 2026

[4] **Multi-Channel Uncertainty-Weighted Score Matching for Conditional Diffusion in Medical UDA**
Chen Li, Meilong Xu, **Xiaoling Hu**, Weimin Lyu, Chao Chen
*European Conference on Computer Vision (**ECCV**)*, 2026

[5] **Act Like a Pathologist: Tissue-Aware Whole Slide Image Reasoning**
Wentao Huang, Weimin Lyu, Peiliang Lou, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Wenchao Han, Ruifeng Guo, Jiawei Zhou, Chao Chen, Chen Wang
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2026

[6] **MATCH: Multi-faceted Adaptive Topo-Consistency for Semi-Supervised Histopathology Segmentation**
Meilong Xu*, **Xiaoling Hu***, Shahira Abousamra, Chen Li, Chao Chen
*Conference on Neural Information Processing Systems (**NeurIPS**)*, 2025
*Short version selected as an **Oral Presentation** at the Imageomics Workshop at NeurIPS 2025*

[7] ***Learn2Synth*: Learning Optimal Data Synthesis Using Hypergradients for Brain Image Segmentation**
Xiaoling Hu, Xiangrui Zeng, Oula Puonti, Juan Eugenio Iglesias, Bruce Fischl‡, Yaël Balbastre‡
*IEEE/CVF International Conference on Computer Vision (**ICCV**)*, 2025

[8] ***TopoCellGen*: Generating Histopathology Cell Topology with a Diffusion Model**

Meilong Xu, Saumya Gupta, **Xiaoling Hu**, Chen Li, Shahira Abousamra, Dimitris Samaras, Prateek Prasanna, Chao Chen

*IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2025 (**Oral Presentation, top 1% of submissions**)

Short version also presented at CVPR 2025 Workshop GMCV

[9] **Hierarchical Uncertainty Estimation for Learning-based Registration in Neuroimaging**

Xiaoling Hu, Karthik Gopinath, Peirong Liu, Malte Hoffmann, Koen Van Leemput, Oula Puonti‡, Juan Eugenio Iglesias‡

*International Conference on Learning Representations (**ICLR**)*, 2025

[10] **Semi-supervised Segmentation of Histopathology Images with Noise-Aware Topological Consistency**

Meilong Xu, **Xiaoling Hu**, Saumya Gupta, Shahira Abousamra, Chao Chen

*European Conference on Computer Vision (**ECCV**)*, 2024

[11] **Brain-ID: Learning Contrast-agnostic Anatomical Representations for Brain Imaging**

Peirong Liu, Oula Puonti, **Xiaoling Hu**, Daniel C. Alexander, Juan Eugenio Iglesias

*European Conference on Computer Vision (**ECCV**)*, 2024

[12] **Semi-Supervised Contrastive VAE for Disentanglement of Digital Pathology Images**

Mahmudul Hasan, **Xiaoling Hu**, Shahira Abousamra, Prateek Prasanna, Joel Saltz, Chao Chen

*International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**)*, 2024

[13] **Hard Negative Sample Mining for Whole Slide Image Classification**

Wentao Huang, **Xiaoling Hu**, Shahira Abousamra, Prateek Prasanna, Chao Chen

*International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**)*, 2024

[14] **Spatial Diffusion for Cell Layout Generation**

Chen Li, **Xiaoling Hu**, Shahira Abousamra, Meilong Xu, Chao Chen

*International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**)*, 2024

[15] **Anomaly-Guided Weakly Supervised Lesion Segmentation on Retinal OCT Images**

Jiaqi Yang, Nitish Mehta, Gozde Merve Demirci, **Xiaoling Hu**, Meera Ramakrishnan, Mina Naguib, Chao Chen, Chialing Tsai

*Medical Image Analysis (**MedIA**)*, 2024

[16] **Topology-Aware Uncertainty for Image Segmentation**

Saumya Gupta, Yikai Zhang, **Xiaoling Hu**, Prateek Prasanna, Chao Chen

*Conference on Neural Information Processing Systems (**NeurIPS**)*, 2023

Short version is also presented at NeurReps Workshop

[17] **Calibrating Uncertainty for Semi-Supervised Crowd Counting**

Chen Li, **Xiaoling Hu**, Shahira Abousamra, Chao Chen

*IEEE/CVF International Conference on Computer Vision (**ICCV**)*, 2023

[18] **Enhancing Modality-Agnostic Representations via Meta-Learning for Brain Tumor Segmentation**

Aishik Konwer, **Xiaoling Hu**, Joseph Bae, Xuan Xu, Chao Chen, Prateek Prasanna

*IEEE/CVF International Conference on Computer Vision (**ICCV**)*, 2023

[19] **Learning Probabilistic Topological Representations Using Discrete Morse Theory**

Xiaoling Hu, Dimitris Samaras, Chao Chen

*International Conference on Learning Representations (**ICLR**)*, 2023 (**Spotlight Presentation, top 6% of submissions**)

*Short version selected as an **Oral Presentation** at Medical Imaging meets NeurIPS Workshop, 2022*

[20] **Confidence Estimation Using Unlabeled Data**

Chen Li, **Xiaoling Hu**, Chao Chen

*International Conference on Learning Representations (**ICLR**)*, 2023

[21] **Structure-Aware Image Segmentation with Homotopy Warping**

Xiaoling Hu

*Conference on Neural Information Processing Systems (**NeurIPS**)*, 2022

[22] **Learning Topological Interactions for Multi-Class Medical Image Segmentation**

Saumya Gupta*, **Xiaoling Hu***, James Kaan, Michael Jin, Mutshipay Mpoy, Katherine Chung, Gagandeep Singh, Mary Saltz, Tahsin Kurc, Joel Saltz, Apostolos Tassiopoulos, Prateek Prasanna, Chao Chen

*European Conference on Computer Vision (**ECCV**)*, 2022 (**Oral Presentation, top 3% of submissions**)

[23] **Trigger Hunting with a Topological Prior for Trojan Detection**

Xiaoling Hu, Xiao Lin, Michael Cogswell, Yi Yao, Susmit Jha, Chao Chen

*International Conference on Learning Representations (**ICLR**)*, 2022

[24] **A Manifold View of Adversarial Risk**

Wenjia Zhang, Yikai Zhang, **Xiaoling Hu**, Mayank Goswami, Chao Chen, Dimitris Metaxas

*International Conference on Artificial Intelligence and Statistics (**AISTATS**)*, 2022

[25] **A Topology-Attention ConvLSTM Network and Its Application to EM Images**

Jiaqi Yang*, **Xiaoling Hu***, Chao Chen, Chialing Tsai

*International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**)*, 2021

[26] **Topology-Aware Segmentation Using Discrete Morse Theory**

Xiaoling Hu, Yusu Wang, Li Fuxin, Dimitris Samaras, Chao Chen

*International Conference on Learning Representations (**ICLR**)*, 2021 (**Spotlight Presentation, top 6% of submissions**)

[27] **Topology-Preserving Deep Image Segmentation**

Xiaoling Hu, Li Fuxin, Dimitris Samaras, Chao Chen

*Conference on Neural Information Processing Systems (**NeurIPS**)*, 2019

Preprints

[1] **TriSegAgent: 3D Medical Annotation with VLM Agents via Tri-View Context Compression**
Qingqiao Hu*, Jialin Li*, Meilong Xu, Kehan Qi, Yipeng Zhang, Bran Hongwei Li, Klaus Mueller‡, **Xiaoling Hu**‡
Tech Report, 2026

[2] **ORCE: Order-Aware Alignment of Verbalized Confidence in Large Language Models**
Chen Li, **Xiaoling Hu**, Songzhu Zheng, Jiawei Zhou, Chao Chen
Tech Report, 2026

[3] **Topo-R1: Detecting Topological Anomalies via Vision-Language Models**
Meilong Xu, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Xin Yu, Weimin Lyu, Kehan Qi, Dimitris Samaras, Chao Chen
Tech Report, 2026

[4] **Topology-Aware Uncertainty for Efficient Vessel Annotation**
Saumya Gupta, Sumedh Ghavat, Yikai Zhang, **Xiaoling Hu**, Prateek Prasanna, Chao Chen
*IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), under major revision*

[5] **Unrolled Networks are Conditional Probability Flows in MRI Reconstruction**
Kehan Qi, Saumya Gupta, **Xiaoling Hu**, Qingqiao Hu, Weimin Lyu, Chao Chen
Tech Report, 2025

[6] **RB-FT: Rationale-Bootstrapped Fine-Tuning for Video Classification**
Meilong Xu, Di Fu, Jiaxing Zhang, Gong Yu, Jiayu Zheng, **Xiaoling Hu**, Dongdi Zhao, Feiyang Li, Chao Chen, Yong Cao
Tech Report, 2025

[7] **Learning to Upscale 3D Segmentations in Neuroimaging**
Xiaoling Hu, Peirong Liu, Dina Zemlyanker, Jonathan Williams Ramirez, Oula Puonti, Juan Eugenio Iglesias
Tech Report, 2025

Mentoring

- Chen Li (**ICLR'23, ICCV'23, MICCAI'24, ECCV'26**), Ph.D. student at Department of BMI, Stony Brook University → Incoming Postdoc at Stanford University
  Since Fall 2021
- Meilong Xu (**ECCV'24, CVPR'25 Oral Presentation, NeurIPS'25**), Ph.D. student at Department of CS, Stony Brook University
  Since Summer 2023
- Wentao Huang (**MICCAI'24, CVPR'26, TMI'26**), Ph.D. student at Department of CS, Stony Brook University
  Since Summer 2023
- Qingqiao Hu (**WACV'27**), Ph.D. student at Department of CS, Stony Brook University
  Since Fall 2024
- Kehan Qi, Ph.D. student at Department of BMI, Stony Brook University
  Since Fall 2025

- Mahmudul Hasan (**MICCAI'24**), Ph.D. student at Department of CS, Stony Brook University
  Summer 2023 - Summer 2024
- Saumya Gupta (**ECCV'22 Oral Presentation, NeurIPS'23**), Ph.D. student at Department of CS, Stony Brook University
  Fall 2021 - Summer 2023
- Jiaqi Yang (**ISBI'21, MICCAI'21, MedIA'24, JBHI'25**), Ph.D. student at Department of CS, CUNY
  Spring 2020 - Summer 2023
- Luca Drole, Master's student in BME, ETH Zurich
  Spring 2025 - Fall 2025
- John Xie, high school student → University of Michigan
  Summer 2021

Invited Talks

Principled Learning for Medical AI: Structure, Reliability, and Interpretability

- Kahlert School of Computing & SCI Institute, University of Utah — Mar. 2026
- School of Computing, University of Georgia — Mar. 2026
- Department of EECS, University of Tennessee, Knoxville — Feb. 2026
- Department of CS, Missouri S&T — Jan. 2026
- Department of CSE, University of North Texas — Nov. 2025
- HIT Webinar - *Healthcare, Intelligence, Technology* — Aug. 2025

***Learn2Synth*: A Learnable Data Synthesis Strategy for Image Segmentation**

- Nobrainer Seminar, Massachusetts Institute of Technology — Jun. 2024

Deep Structural Reasoning for Biomedical Imaging

- School of CAI, Arizona State University — Feb. 2024

Topology-Aware Deep Image Segmentation

- MICCAI'23 tutorial on *Topology-Driven Image Analysis*, Vancouver — Oct. 2023
- Geometry and Topology meet Data Analysis and Machine Learning — Aug. 2021

Learning Topological Representations for Deep Image Understanding

- Department of CS, Florida State University — Apr. 2023
- Department of BMI, Ohio State University — Mar. 2023
- Department of CS, Rochester Institute of Technology — Feb. 2023
- Department of ECE, University of California, Riverside — Feb. 2023
- Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital & Harvard Medical School — Nov. 2022

Learning Probabilistic Topological Representations Using Discrete Morse Theory

- Medical Imaging meets NeurIPS Workshop, New Orleans — Dec. 2022

Topology-Informed Image Analysis

- Center for Computational Neuroscience, Flatiron Institute — Oct. 2022

Topology-Aware Segmentation Using Discrete Morse Theory

- International Conference on Learning Representations (ICLR) — May 2021

Industry Experience

- **Allen Institute, USA** — May 2022 - Aug. 2022
  Research Intern
  Mentor: *Dr.* Matheus Viana
  Topic: Topology-Aware Image Segmentation
- **United Imaging Intelligence (UII), USA** — May 2021 - Aug. 2021
  Research Intern
  Mentor: *Dr.* Shanhui Sun
  Topic: Deep Shape Model Based Network
- **Tencent Youtu Lab, China** — Jun. 2017 - Jan. 2018
  Research Intern
  Mentor: *Dr.* Yuwing Tai
  Topic: Object Detection and Clothes Search